

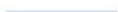
Embeddings, RAG and Guardrails

Build grounded LLM experiences with measurable quality and safeguards.



Learning outcomes

01  Build retrieval over a controlled document set.

02  Trace answers to evidence.

03  Apply access, safety, and fallback controls.

The concept system

01

embeddings

02

retrieval

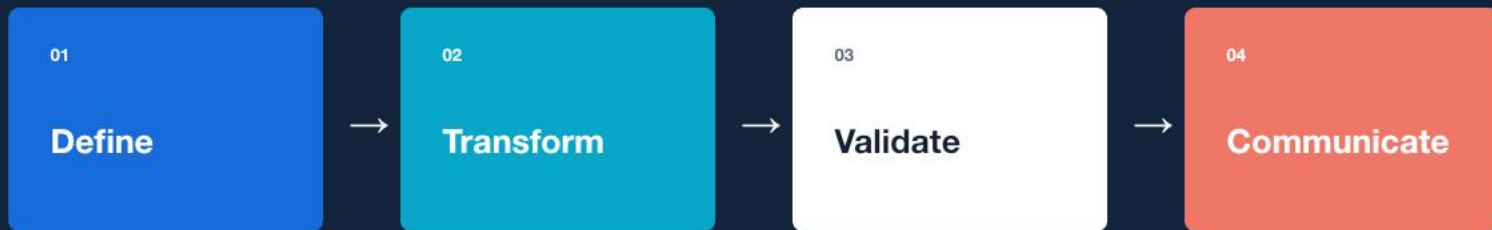
03

grounding

04

guardrails

A repeatable reasoning loop



Build a cited assistant over course policies and notes.

Guided lab

LAB BRIEF

**Build a cited assistant over
course policies and notes.**

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A RAG prototype with retrieval evaluation and threat model.

QUALITY GATE

- **Reproducible**
- **Tested**
- **Decision-relevant**

WEEKLY ASSESSMENT

Grounded Knowledge Assistant

Retrieval 25%, grounding 25%, evaluation 20%, safeguards 20%, documentation 10%.

What should remain after the lesson?

CONCEPT

embeddings + retrieval

PRACTICE

Build a cited assistant over course policies and notes.

EVIDENCE

A RAG prototype with retrieval evaluation and threat model.
